Subgradients and the Subdifferential

$f : \mathbb{R}^n \to \mathbb{R}$

g is a subgradient of f at x : $g \in \partial f(x)$

iff : $f(y) \geq f(x) + g^T(y-x) \quad \forall y.$

If f has a gradient at x, then $\partial f(x) = \{\nabla f(x)\}$

Optimality Conditions

$\boxed{\min: f(x)}$ $\boxed{f \text{ convex}}$

If f diff'ble at $\hat{x}$: $\hat{x}$ opt. iff $\nabla f(\hat{x}) = 0$

For any convex f : $\hat{x}$ opt. iff $0 \in \partial f(\hat{x})$

Note: $0 \in \partial f(\hat{x})$: $f(y) \geq f(\hat{x}) + 0^T(y - \hat{x}) = f(\hat{x}) \quad \forall y$

i.e., $f(\hat{x}) \leq f(y) \quad \forall y.$

<u>Constrained Opt</u>:

$\min: f_0(x)$

st: $f_i(x) \leq 0$

$h_j(x) = 0$

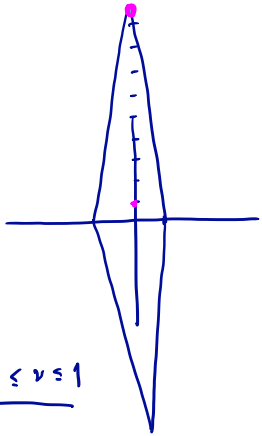
<u>KKT</u>: $\hat{x}$ feasible, $\hat{\lambda}, \hat{\nu}$ feasible, CS } no grad needed to define.

stat: $\underbrace{0 = \nabla f_0(\hat{x})}_{0 \in \partial f_0(\hat{x})} + \sum \lambda_i \nabla f_i(\hat{x}) \quad \sum \nu_j \nabla h_j(\hat{x})$

Not Necessarily a Descent Direction

$$\boxed{f(x - \eta g) \le f(x)}$$ — not nec. true.

comp. to:



Ex: $f(x_1, x_2) = 10 \cdot |x_1| + \underline{1 \cdot |x_2|}$

Sub diff at $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

$$\partial f \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \cdot v \\ 1 \end{pmatrix}, \quad v \in \partial |x_1| \Big|_0 : \underline{-1 \le v \le 1}$$

$$f\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} - \eta \begin{pmatrix} 10 \\ 1 \end{pmatrix}\right) = f\begin{pmatrix} -10 \cdot \eta \\ 1 - \eta \end{pmatrix} = 100 \cdot \eta + (1 - \eta) = 1 + 99 \cdot \eta > 1 = f\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Nevertheless: subgradients can be used like grad, to build min alg.